A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing a ground-truth budget into the weak labels helps only modestly — and *where* and *how* it is spent show little effect.

A consistent account: the student largely reproduces the teacher's errors, and ground truth corrects mainly the examples it directly labels — so the effective lever is *volume*, not placement or combination.

GPT-2 family · BoolQ + SciQ · 3 seeds · paired per-(pair,seed) contrasts · pre-registered

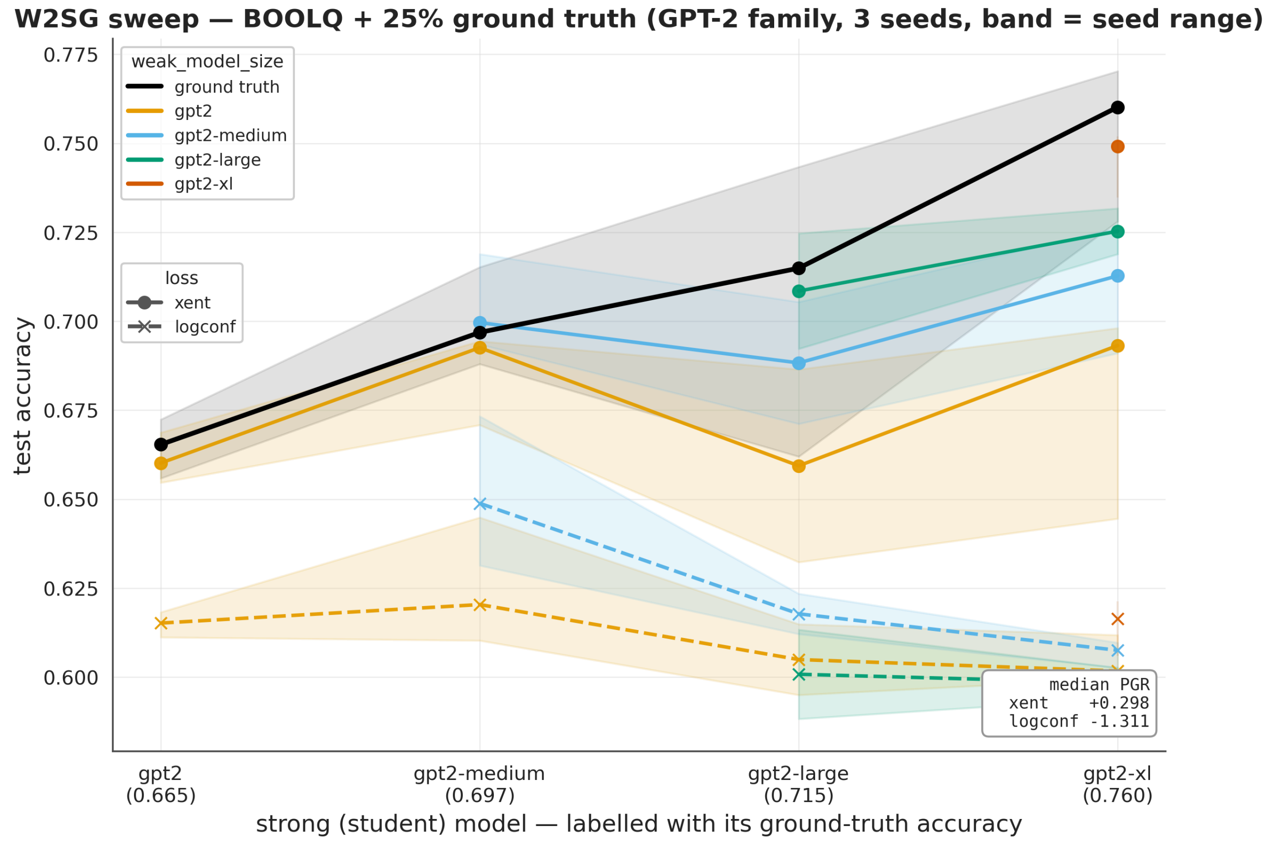
Scope & setup

- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher*'s labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student–teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).

3 seeds · paired per-(pair,seed) contrasts · pre-registered · gpt2-large (seed 1) excluded by rule

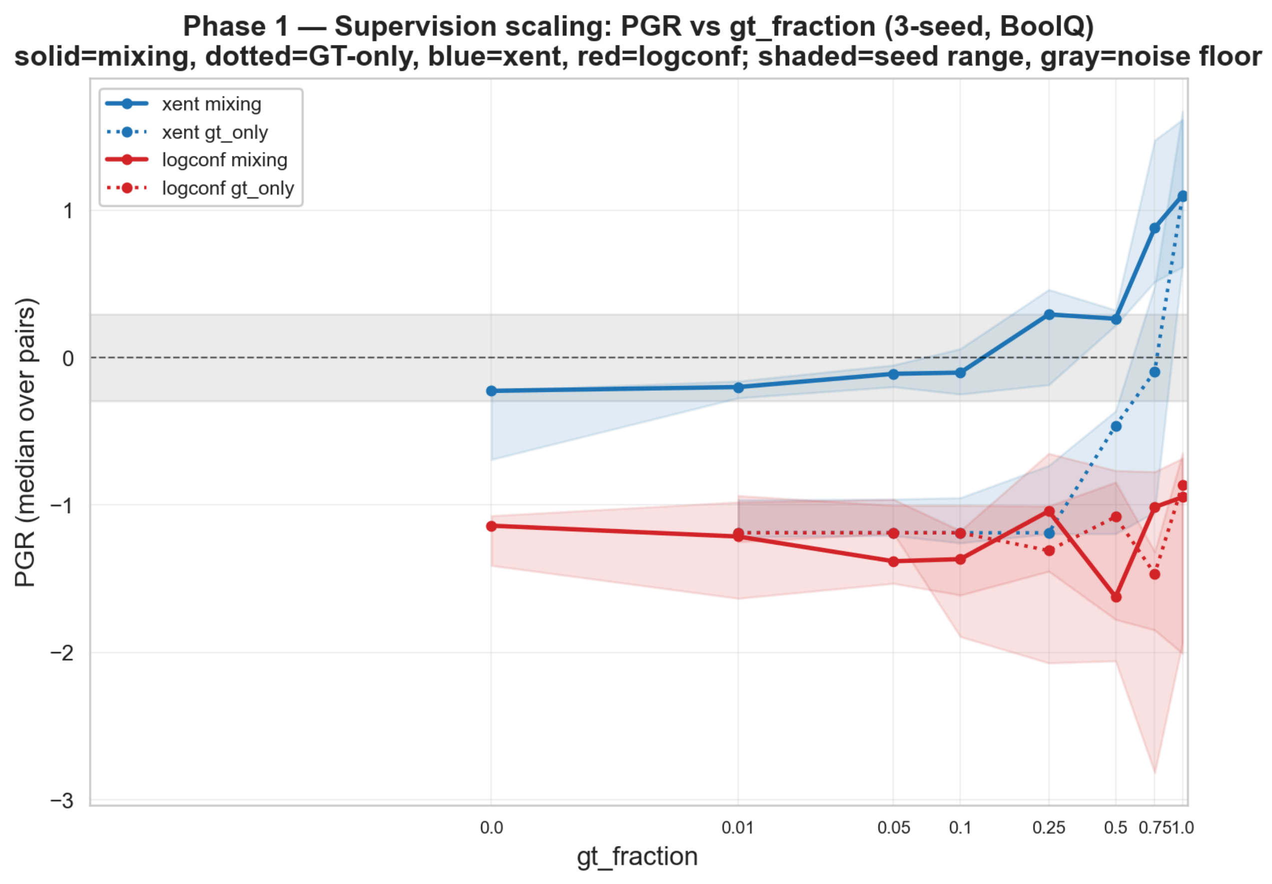
GPT-2 sweep at 25% ground truth

- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — standard sweep format.
- **Median PGR (xent) = +0.30** across the sweep; logconf (dashed) sits well below xent.
- The 0% baseline is **−0.27**; 25% GT moves the median to **+0.30** — a small positive.



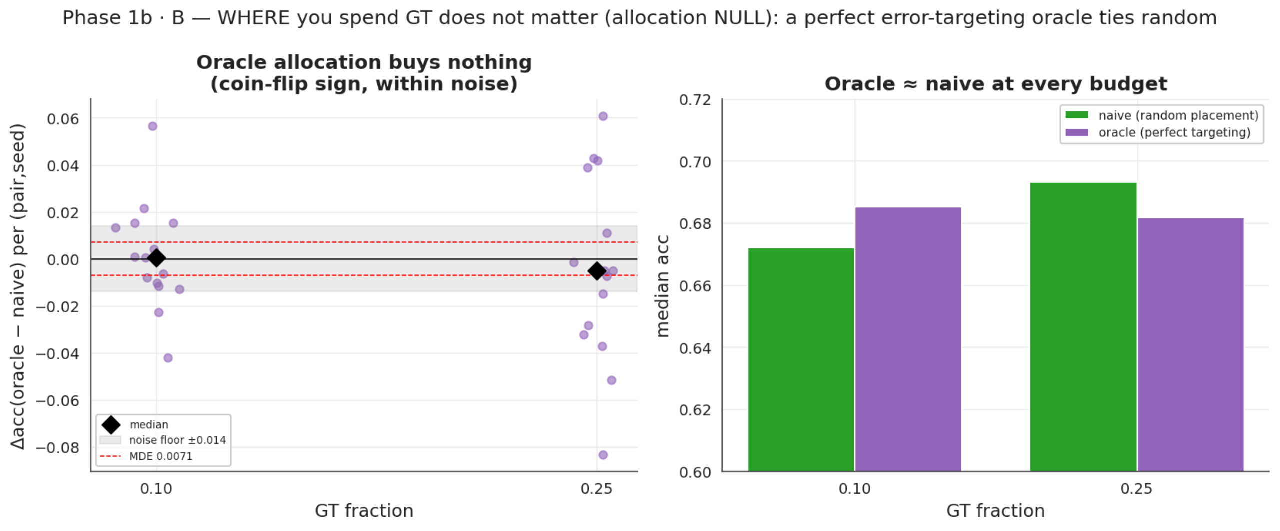
How *much*? — back-loaded, and saturating by ~75%

- Up to 10% GT, the gain over the 0% baseline stays within the 0.014 noise floor.
- Median xent PGR is back-loaded: −0.22 → +0.30 (0.25) → +0.28 (0.50) → +0.90 (0.75) → +1.04 (1.0).
- The 0.75 point: the 0.50→0.75 step is the largest, and 0.75→1.0 is within noise — the gain is essentially complete by ~75%.
- A pre-registered concave-knee-at-25% prediction was refuted, and retracted after the multi-seed data.



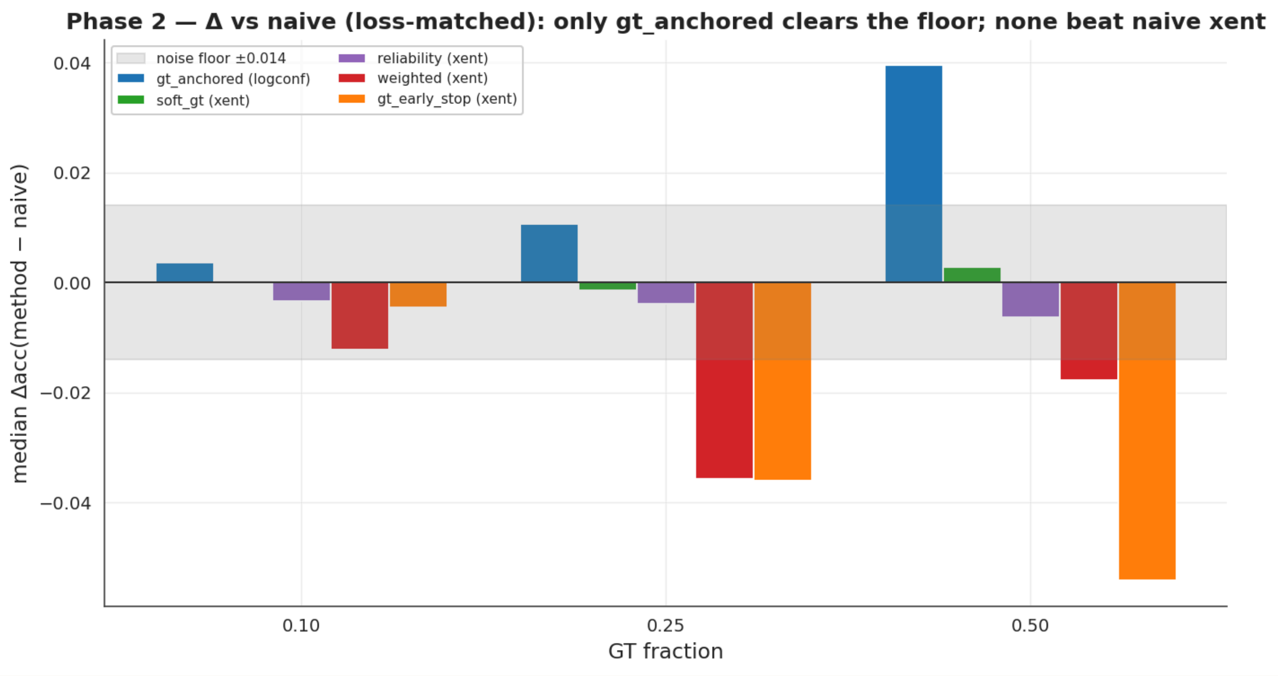
Where it's spent — within noise (allocation null)

- An **oracle** uses held-out GT to place the budget exactly on the **teacher-wrong rows** — an upper bound on any allocation rule.
- Paired against random placement at matched budget: oracle − random = **+0.0006** (0.10), **−0.0049** (0.25) — both within the **MDE (0.0071)**, so a genuine null, not underpowered.
- If the best possible placement ties random, allocation heuristics have **little to gain** on this testbed.



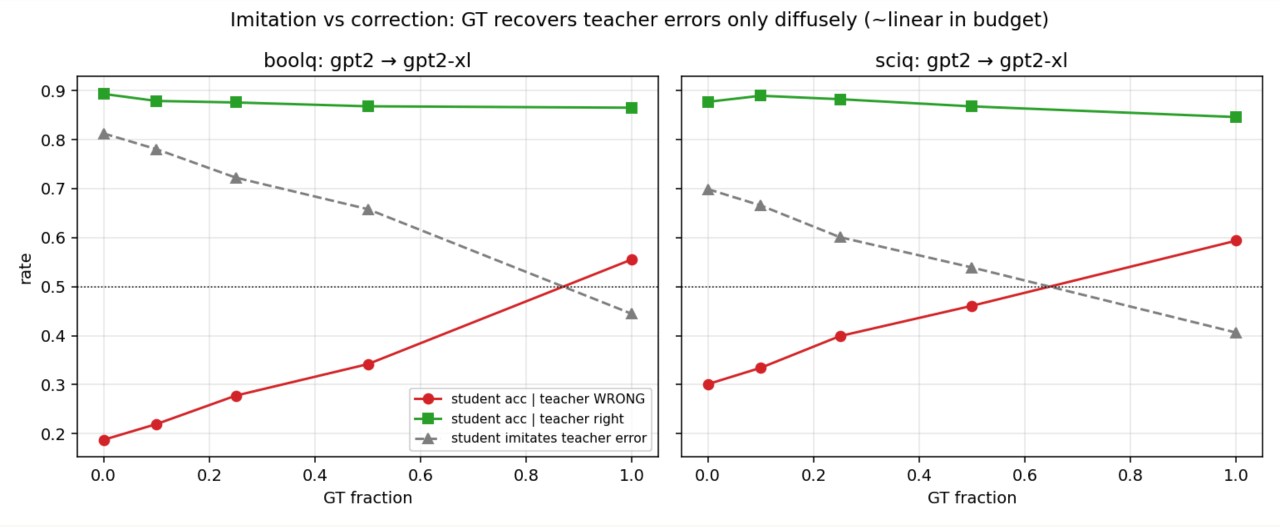
How it's combined — within noise (combination null)

- Five methods vs naive mixing at matched budget: M1 GT up-weighting · M2 soft-GT targets · M3 GT-anchored logconf · M4 reliability-weighted weak labels · M5 GT-based early stopping.
- Median Δ vs naive ≈ 0 across {0.10, 0.25, 0.50}; none shifts the curve left or raises the ceiling.
- M3 (gt-anchored) is the only method above the floor (+0.040 at 0.50), but it recovers logconf toward xent without exceeding plain xent (0.642 vs 0.697).



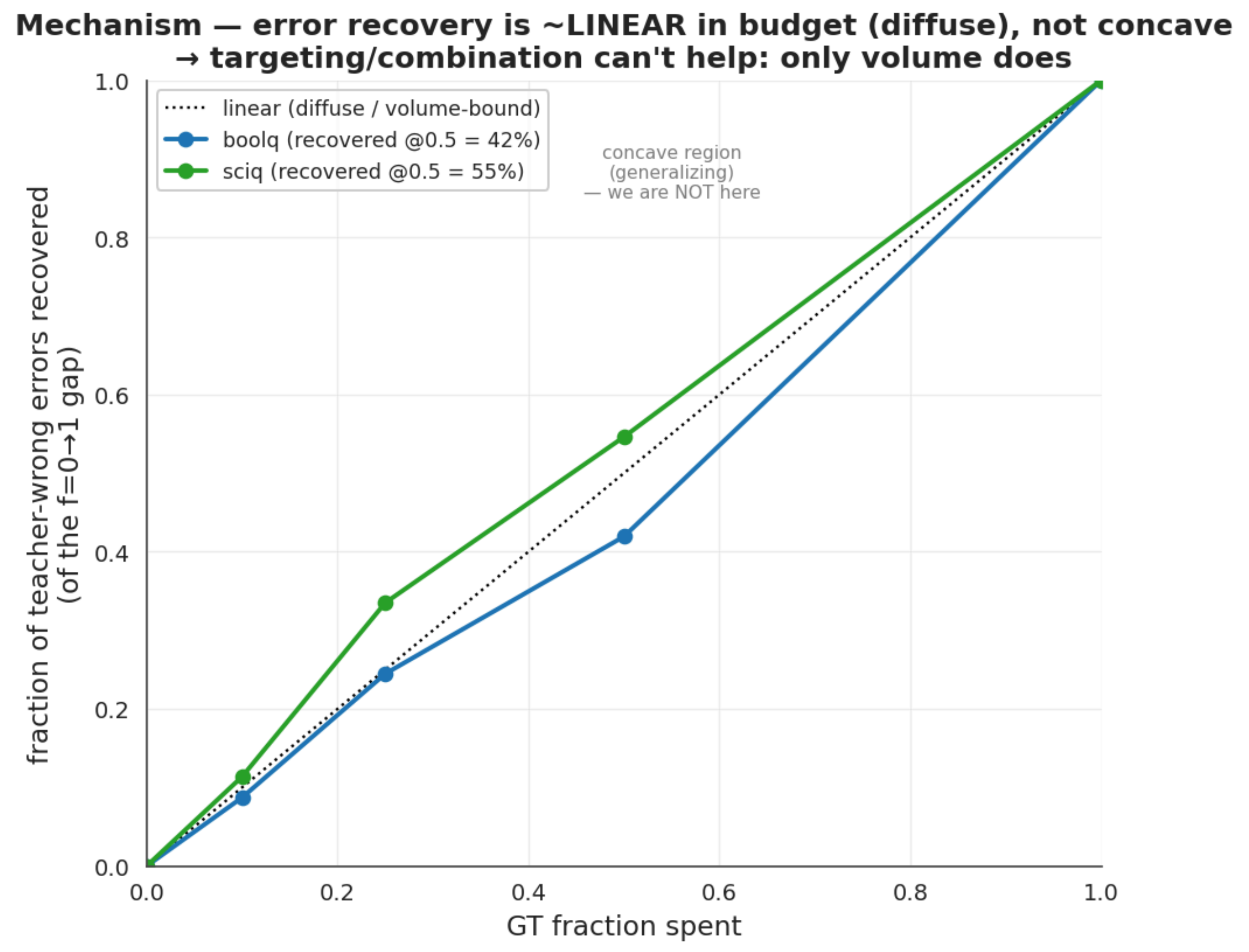
Mechanism: the student reproduces the teacher's errors

- Probe: gpt2 → gpt2-xl, joining teacher and student per-example test predictions.
- On teacher-wrong rows at 0% GT, the student reproduces the teacher's wrong answer 81% (BoolQ) / 70% (SciQ) of the time.
- Errors are largely inherited rather than independent — so the rows needing correction are identifiable in principle.



Mechanism: recovery is ~linear in budget (volume-bound)

- Recovery of teacher-wrong rows is **~linear in budget** — 42% / 55% of the 0→100% gap recovered at 50%, tracking $y = x$ rather than a concave curve.
- A GT label's marginal value appears **roughly independent of which row it lands on**.
- Under volume-bound recovery, placement (*where*) and re-weighting (*how*) have **little leverage** — consistent with both nulls.



Pre-registration scorecard — hits and misses

Phase 1 (P1–P6, git-anchored before seeds 0/2 existed)

	Prediction	Outcome
P1	knee at ~25%	✗ refuted — back-loaded, no knee (retracted)
P2	≤10% GT inert	✓ within noise
P3	mixing > GT-only	✓ (confound named → resolved in 1b)
P4	logconf null	✓ inert at every budget
P5	scale interaction (gap → more GT)	— underpowered at GPT-2 scale
P6	0.25–0.50 plateau	✓ flat in that range

Phase 2 (M1–M5, registered before the portfolio ran): M2 ✓ null, M4 ✓ null (as flagged uncertain), M3 ◑ (strongest bet — rescues logconf, still < xent), M1/M5 ✗ negative.

Experiments across the three axes

Axis	Experiment	Outcome
How much	8-point fraction sweep, 0 → 1	back-loaded, saturates ~0.75
Where	error-targeting oracle + random control	null — within MDE
How	5 combination / loss methods	null — 1 above floor, still < xent
Loss	xent vs logconf	logconf inert → dropped
Tasks	BoolQ + SciQ	replicates on both
Mechanism	imitation-vs-correction probe	recovery ~linear in budget
Robustness	8-seed variance; 5-seed baseline pass	exclusion by rule; 3-seed conclusions hold

Coverage spans the three axes plus loss, task, and robustness checks; the negatives are concordant with the volume-bound account.

Takeaways

- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- Decomposed the budget question into *how much / where / how* — a powered null on the last two, and a back-loaded, saturating curve on the first.
- One account — **inherited errors plus volume-bound recovery** — is consistent with all three.
- So the binding constraint here looks like **scale**, not allocation or combination — which sets the next experiments.

Next steps

1. **Vary the capability gap.** Volume-bound recovery predicts a GT label's value is gap-invariant. Re-measure recovery under wider student–teacher gaps (early-checkpoint / label-noised teachers; larger families if scope allowed) — a shift from linear to **concave** would mean targeted GT generalizes, reopening *where* and *how*.
2. **Test whether teacher errors are structured.** Probe the student's hidden states for teacher-correctness from held-out GT. Low AUC ⇒ the wrong rows are diffuse, not separable for the budget to exploit — the mechanism under the allocation null. Near-free.
3. **Schedule the budget over training.** The mixture is held stationary; ordering is unexamined. Compare GT-first / annealed / interleaved at matched budget — refinement-scheduling work suggests order can outweigh the loss objective under weak supervision.
4. **Iterate the weak-to-strong loop.** Re-derive the teacher from the budget-trained student, repeat at fixed total GT — does a one-pass-null budget compound across rounds (single-round elicitation vs label information)?

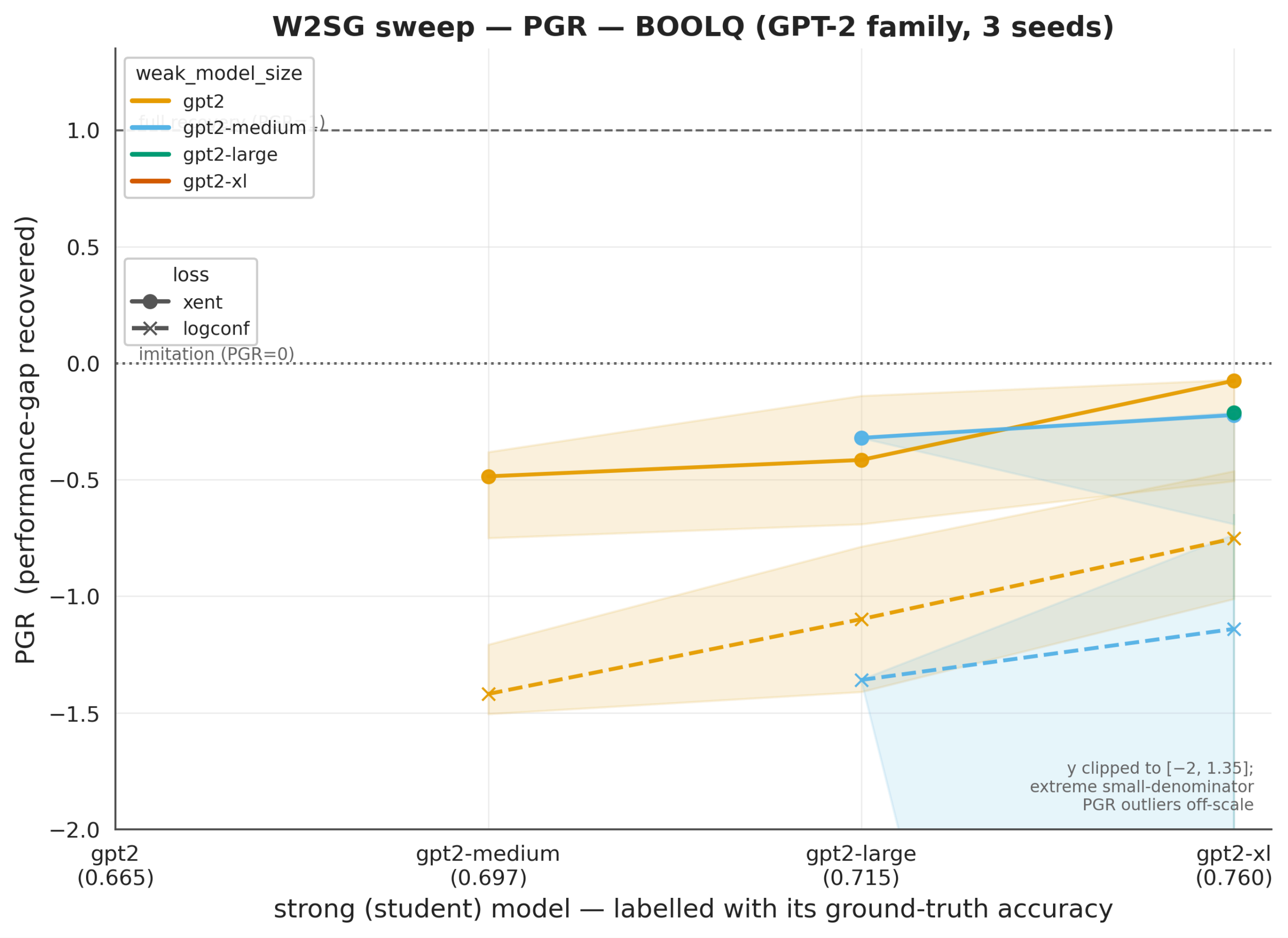
Ordered by bearing on the volume-bound account: (1) discriminates it · (2) tests its premise · (3–4) probe axes the budget framing leaves open.

Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

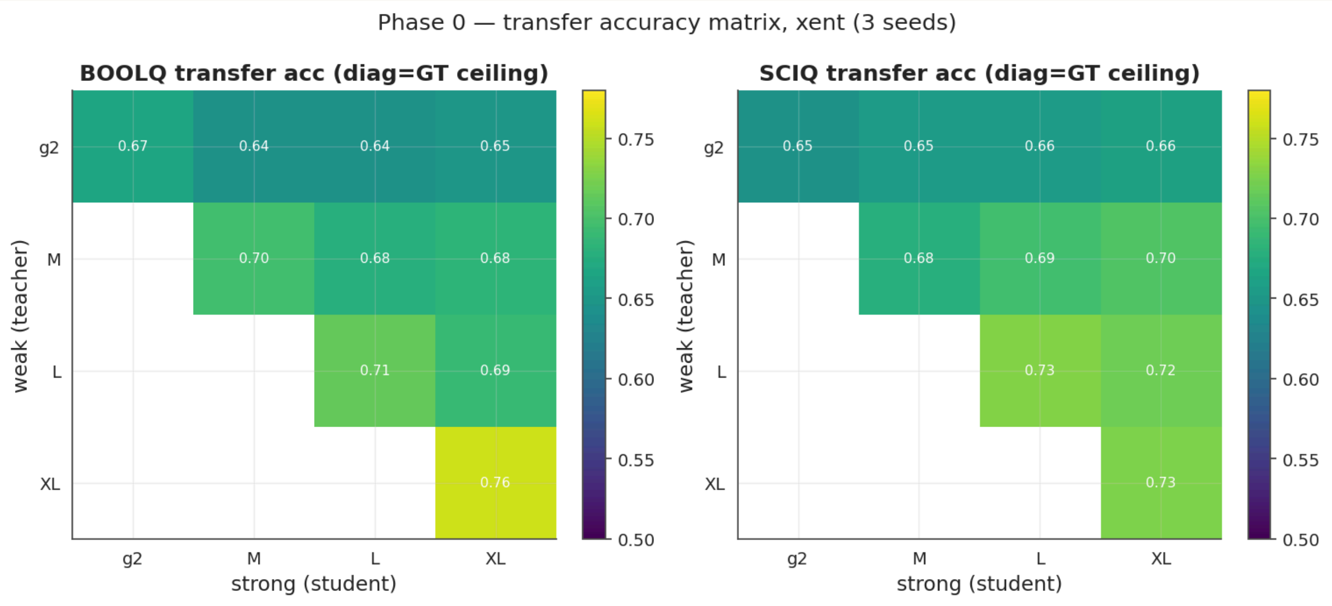
Appendix — frac=0 reproduction

- Standard W2SG sweep, GPT-2 family, BoolQ, 0% GT, PGR axis.
- Median xent PGR −0.27; several pairs fall below the imitation line — BoolQ is low-signal at 0% GT.
- This is the baseline the 25%-GT sweep is measured against.



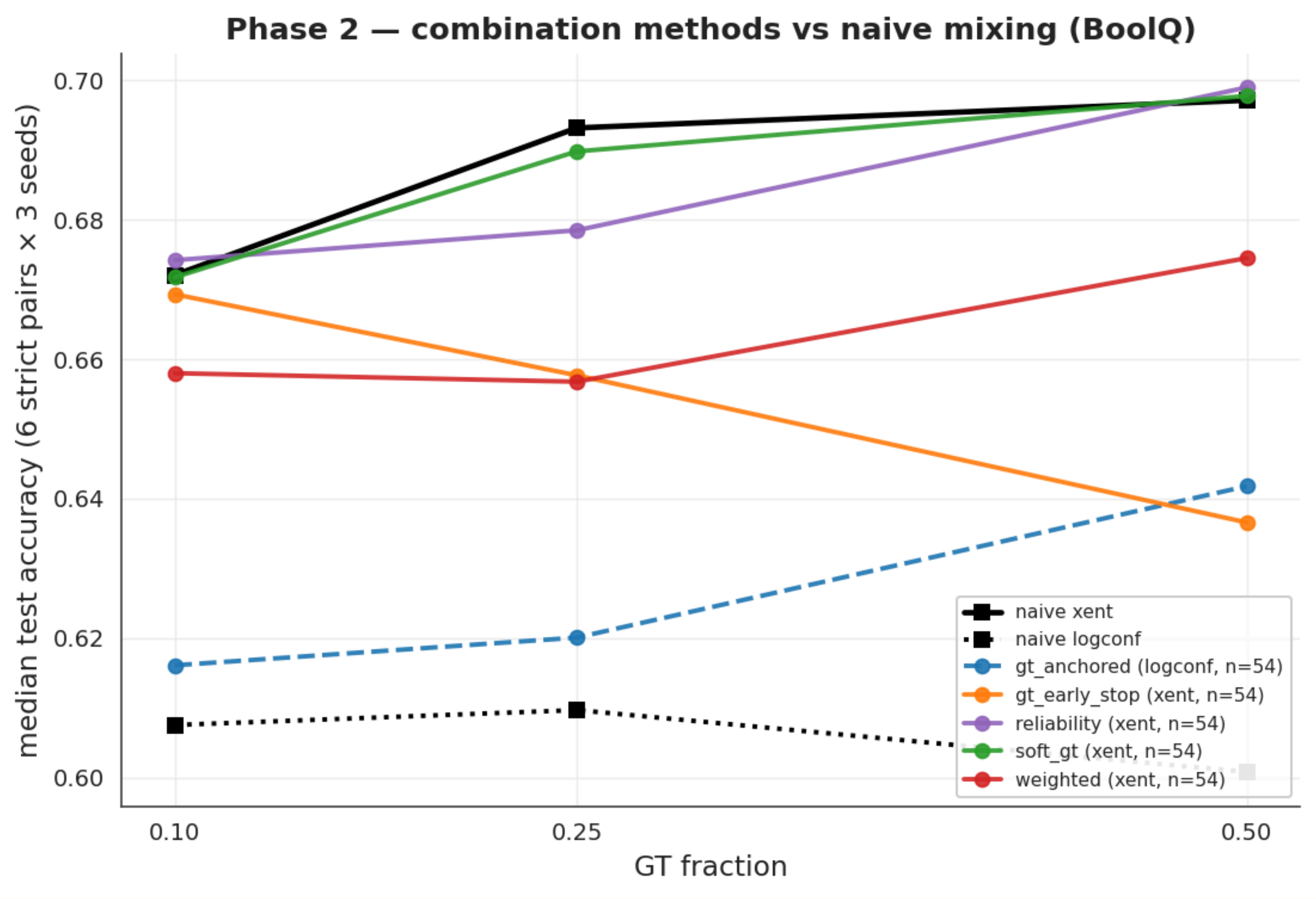
Appendix — transfer tracks the teacher more than student capacity

- With the teacher fixed, scaling the student adds ~+0.003; with the student fixed, a better teacher adds ~+0.04.
- Transfer tracks the teacher far more than student capacity — the static counterpart of the imitation result.



Appendix — combination portfolio, per-method detail

- All five methods track naive mixing across {0.10, 0.25, 0.50}.
- gt-anchored (logconf) is the only method above the floor; it recovers logconf but stays under xent.



Appendix — weak labels carry information (de-confound)

- Ordering: random < gt_only < naive mixing (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise reduces accuracy → the mixing gain reflects weak-label information, not just added rows.